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DIBRIS

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RESEARCH TRACK II

Final Assignment

**How Safe Navigation Degrades Under Sensor and Actuator Faults: A
Comparison Between Three Nav2 Local Planners**

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1 Introduction

The deployment of autonomous ground robots in semi-structured environments heavily relies on their ability to navigate safely and efficiently. However, operational safety is constantly threatened by hardware and software malfunctions. Continuous degradation in sensors and actuators due to environmental interference, wear, or component aging can severely compromise the robot’s reliability and its real-time decision-making [1]. On a ground robot the channels most exposed are the LiDAR and the wheel motors.

Obstacle avoidance is usually seen as the main concern under such conditions, but we observed that focusing only on collisions misses part of the problem. Under some sensor or actuator faults, mobile robots often fail without ever touching an obstacle: instead they freeze or get stuck in open space. To capture the whole range of fault-induced failures, we therefore frame the study around safe navigation more broadly.

In the Robot Operating System (ROS2) ecosystem, the Navigation2 (Nav2) stack has a modular architecture that lets us swap local trajectory planners easily [2]. A recent benchmark compares these controllers under perception-side and environmental stressors [2], but how they compare in resilience to a range of continuous sensor and actuator faults (especially per-wheel actuator degradation) measured on both safety and liveness, is still largely unexplored. To address this gap, we pose the following research question: How does safe autonomous navigation in ground robots degrade under continuous sensor and actuator faults, and how do standard local planners (DWA, RPP, and MPPI) differ in their robustness?

To investigate this, we present a reproducible, simulated benchmarking framework. Our main contributions are as follows:

- A factorial benchmark of 306 trials comparing three Nav2 local planners under four continuous fault models (sensor bias, drift, noise, and actuator degradation) on a common severity scale.
- A metric scheme that separates perception-safety from liveness-control, so each fault is judged on the axis it actually degrades.
- The identification of four distinct failure signatures: margin erosion, late collision, hyper-conservative freezing, and open-space entrapment.
- A clear robustness ranking under actuator degradation ($DWA \geq RPP \gg MPPI$), the regime where the planners differ most.

2 Literature Analysis

2.1 Fault Taxonomy and Injection

Evaluating robustness first needs a clear definition of failure modes. Nasri et al. [1] built a comprehensive fault taxonomy for safety-critical autonomous vehicles within the ISO 26262 functional safety framework. They group continuous faults into sensor bias, sensor drift, sensor noise, and actuator degradation, and stress single-active-fault modeling to isolate hardware-specific degradation. Their vehicle is car-like and their fault magnitudes come from full-scale automotive data, but we reuse the same taxonomy for a differential-drive ground robot. Because road-derived magnitudes carry no meaning for a robot that turns in place inside a small indoor arena, we set the fault magnitudes from the geometry of our own robot instead of from automotive values.

2.2 Benchmarking Nav2 Local Planners

The Nav2 stack provides a standard architecture for mobile robot navigation. Urrea and Valencia-Aragón [2] systematically compared three local controllers from different control-theoretic families: DWA (Dynamic Window Approach), RPP (Regulated Pure Pursuit), and MPPI (Model Predictive Path Integral). Their benchmark, run across several environmental stressors (e.g., slope-induced costmap distortions and narrow corridors), supports using this trio as a standard baseline for Nav2.

2.3 Robustness Metrics in Embodied Navigation

Evaluating navigation under stress needs metrics that capture both the outcome (success/failure) and the quality of the path. In the embodied-AI domain, Chattopadhyay et al. [3] use the Success Rate together with Success Weighted by Path Length (SPL), a normalized metric that balances task completion with path efficiency, and also describe failure cases through the closest distance the agent reaches to the target. We adopt SPL and add the final distance to the goal, which, unlike SPL, stays informative even when an episode

fails. We also use the stopped-time ratio defined by Urrea and Valencia-Aragón [2] to detect hyper-conservative “freeze-like” behaviors and add the minimum clearance between the robot footprint and the nearest obstacle. Together these cover both the safety and the liveness of the robot.

2.4 Controller Mechanisms and Model Dependency

The differences in planner robustness come from their algorithms. Wei et al. [4] note that DWA relies on real-time velocity sampling scored against an instantaneous costmap, while MPPI uses stochastic optimal control, rolling out sampled trajectories based on a specific kinematic or dynamic model of the vehicle. Macenski et al. [5], by contrast, describe RPP as a purely geometric path-following algorithm, stating explicitly that “vehicle dynamics are not modeled in this style of path tracking algorithm.” We argue that this split, model-based (MPPI) versus model-free (RPP) or short-horizon sampling (DWA), is the key factor in how well a planner tolerates model-mismatch, such as the per-wheel loss of effectiveness (LOE) adopted for differential-drive robots by Ayyildiz and Tilki [6].

2.5 Identified Gap

Despite extensive navigation research, a gap remains: to the best of our knowledge, no study has yet combined a continuous, multi-dimensional fault taxonomy (both sensor and actuator degradation) with a standard Nav2 local planner benchmark evaluated jointly on safety and liveness. Existing benchmarks usually rely on perception-side stressors or environmental constraints rather than per-wheel actuator faults, leaving the link between controller mechanisms and continuous hardware degradation unexplored.

3 Methodological Proposal

Our proposal adapts the continuous-fault taxonomy of Nasri et al. [1] (originally developed for car-like autonomous vehicles) to a ROS2 ground robot, to study how these faults degrade the reactive safe-navigation behavior of three Nav2 local planners. Following a single-active-fault approach, each trial injects exactly one fault, switched on only once the robot starts moving, so the costmap and the initial global plan are built from clean sensor readings first; this gives a clean per-fault baseline, as in [1].

To get a comparable dose-response across different faults, we put them all on a single dimensionless severity scale, given as a percentage of a per-fault maximum. This maximum does not come from automotive ranges; instead we anchor it to our own system: the robot and costmap geometry for the sensor faults, and to the baseline planner’s failure onset for the actuator. We set the highest level at the onset of failure (observable but not catastrophic), so the severity levels concentrate where the planners are most likely to separate.

A further premise is that no single number can fairly summarize all four faults, because they degrade different aspects of navigation. Sensor bias and drift erode the geometric safety margin, whereas noise and actuator faults never lead to a collision but reduce the robot’s ability to make progress. We therefore group the faults into two families (perception-safety and liveness-control) and give each a primary metric matched to its failure signature (minimum clearance and a stopped-time ratio, respectively), while keeping success rate and SPL as cross-cutting measures.

Finally, so that any degradation we see comes from reactive obstacle avoidance and not from a corrupted pose estimate, we give the navigation stack perfect localization: the robot’s true pose feeds localization, while the fault acts only on the perception and actuation channels. This isolates the avoidance behavior at the cost of realism, a trade-off we revisit in the Discussion.

3.1 Hypotheses

The study is organized around three falsifiable hypotheses, each with a fixed test, direction, and multiplicity correction:

- **H1 (dose-response):** increasing fault severity degrades navigation performance, assessed per planner and fault on the family-primary metric and on SPL.
- **H2 (planner robustness):** the three planners differ in robustness, compared on the cross-cutting SPL metric and paired by spawn to remove spatial variance.
- **H3 (failure signatures):** the four faults produce qualitatively distinct failure modes; because the two families measure different, non-comparable axes, H3 is treated descriptively rather than as a single omnibus test.

4 Experiment Design

4.1 Simulation Setup and Navigation Stack

Experiments were conducted using a simulated TurtleBot3 Waffle (a differential-drive platform) running ROS2 Jazzy and the Nav2 stack within Gazebo. The environment is a 10x10 m enclosed arena bounded by perimeter walls and containing a 4x4 regular grid of cylindrical obstacles.

For a fair comparison, all three local planners (DWA, RPP, and MPPI) share identical global planning, costmap, and behavior server configurations; the local trajectory planner plugin is the only configuration difference between them. We also capped the maximum velocity at 0.26 m/s for all controllers, so that differences in nominal speed do not confound the time and path-efficiency metrics.

We implement the perfect-localization choice motivated earlier as follows: a dedicated node publishes the `map→odom` transform from the true Gazebo pose, while AMCL is disabled. The perimeter walls are given to the global costmap as prior knowledge, while the internal obstacles are discovered live and only through the fault-corrupted LiDAR. All performance metrics are computed against the exact ground-truth odometry (`/odom_truth`) exported directly from the Gazebo physics engine.

4.2 Fault Injection Architecture

Faults were injected at runtime via a transparent passthrough node intercepting the LiDAR scans (`/scan` to `/scan_faulty`) and the base controller commands (`/cmd_vel_raw` to `/cmd_vel`), with the single active fault armed upon the robot's first motion.

- **Sensor Bias:** A constant offset b added to all valid LiDAR ranges ($\tilde{r} = r + b$).
- **Sensor Drift:** A time-varying ramp offset $\tilde{r} = r + \alpha(t - t_0)$, where α is the drift rate.
- **Sensor Noise:** Additive Gaussian interference $\mathcal{N}(0, \sigma^2)$ sampled independently per beam.
- **Actuator Degradation:** Modeled as a multiplicative Loss of Effectiveness (LOE) $\eta \in (0, 1]$ applied exclusively to the right wheel ($\tilde{v}_R = \eta \cdot v_R$).

4.3 Severity Scale and Fault Magnitudes

All four faults share one severity scale $s \in \{0, 0.25, 0.50, 0.75, 1.0\}$, reported throughout as $\{0, 25, 50, 75, 100\}\%$ ($s = 0$ being the nominal condition). The physical magnitude is $s \cdot S_{max}$ for the sensor faults and $\eta = 1 - 0.60s$ for the actuator. The sensor-fault levels S_{max} are anchored to the robot and costmap geometry (`robot_radius` = 0.22 m, `inflation_radius` = 0.70 m) and validated at the tightest spawn. Table 1 lists the resulting magnitudes.

Table 1: Fault magnitudes per severity level.

Fault	Unit	25%	50%	75%	100%
Sensor bias	b [m]	0.30	0.60	0.90	1.20
Sensor drift	α [m/s]	0.010	0.020	0.030	0.040
Sensor noise	σ [m]	0.10	0.20	0.30	0.40
Actuator	η [-]	0.85	0.70	0.55	0.40

4.4 Factorial Design and Evaluation Metrics

The design crosses 3 planners $\times$ 4 fault types $\times$ 4 severity levels $\times$ 6 fixed spawn locations for the fault-on trials (288), augmented by a shared nominal baseline of 3 planners $\times$ 6 spawns (18), yielding a total of 306 trials. The target goal is fixed. The 6 spawn points act as a replicated spatial blocking factor, enabling paired statistical comparisons across planners.

An episode is a success if and only if Nav2 reports the goal as reached, the robot halts within 0.30 m of the goal, and no collision occurs. A collision is defined as the geometric footprint clearance dropping to zero or below, which triggers an immediate task abort. Performance is evaluated with a family-primary metric (minimum clearance for the perception-safety faults, the stopped-time ratio for the liveness-control faults) complemented by the cross-cutting measures below:

- **Success Rate:** the fraction of successful episodes, used as a descriptive dose-response indicator.
- **Success Weighted by Path Length (SPL):** employed in the embodied-AI domain [3]. The optimal path length for SPL is computed offline via A* on the occupancy grid inflated by the robot radius.

- Stopped-Time Ratio (R_{stop}): defined in [2] to detect “freeze-like” behaviors, it measures the fraction of the trial spent below $v_{stop} = 0.03$ m/s. We adapt this metric by normalizing over the total trial duration (T_{trial}) rather than the time-to-goal, ensuring it remains mathematically defined during entrapment or freezing failures.
- Minimum Clearance: a direct geometric measurement of the safety margin between the robot’s footprint and the nearest obstacle.

4.5 Statistical Analysis Methodology

Because the sample size per cell is small ($n = 6$) and the metrics are non-Gaussian (bounded, or including failures), we adopted the non-parametric statistical methodology of the reference benchmark [2], with a significance level of $\alpha = 0.05$ (two-tailed). Family-wise error over multiple comparisons was controlled with the Holm-Bonferroni step-down method [2]. Dose-response degradation (H1) was assessed with the Kruskal-Wallis test (the one-factor counterpart of the rank-based factorial test used in [2]), reporting the $\epsilon^2 = H/(N - 1)$ effect size, and the magnitude of planner differentiation with Cliff’s δ and the median difference, as in [2]. Planner robustness (H2) was evaluated on SPL with the Wilcoxon signed-rank paired test: whereas [2] applies unpaired rank-based comparisons, we use a paired test to exploit the spawn blocking, pairing observations by spawn across the non-zero severities ($n = 24$ per fault) to absorb spatial variance.

5 Results and Visualization

5.1 Aggregate Performance Overview

Across the 306 trials, the global Success Rate (SR) is 0.833. Looking only at the fault-on trials (excluding the shared nominal baseline) gives a clear pooled robustness ranking: DWA (0.896) exceeds RPP (0.812), which in turn exceeds MPPI (0.760). However, breaking the results down by fault type shows that the degradation is not spread evenly. Perception faults (bias, drift, noise) degrade the shared sensory stack, giving comparable SR drops across all planners. The divergence between planners, in contrast, comes almost entirely from actuator degradation, which acts directly on the controller loop.

5.2 H1: Dose-Response Degradation

Fig. 1 shows that each fault degrades navigation more and more as its severity grows, with a broadly monotonic trend. The effect of severity is statistically significant in all twelve planner fault combinations, both on each family’s primary metric and on the cross-cutting SPL (Kruskal-Wallis test, all $p < 0.05$ after Holm correction), with large effect sizes ($\epsilon^2 \in [0.48, 0.98]$). The injected faults therefore drive the robot away from clean navigation consistently and strongly, which is the premise the rest of the analysis rests on.

As Fig. 1 shows, the two fault families degrade different axes. Sensor bias gradually erodes the minimum clearance, reducing the safety margin to near-collision levels (~ 0.055 m) at 100% severity without ever colliding, whereas sensor drift drives the clearance to near-zero and produces actual contacts. Sensor noise and actuator degradation, in contrast, never produce a collision (the clearance stays wide under noise, and narrows for DWA and RPP only at the top actuator levels) but cause a sharp rise in the stopped-time ratio (R_{stop}), showing up as hyper-conservative freezing or entrapment in open space.

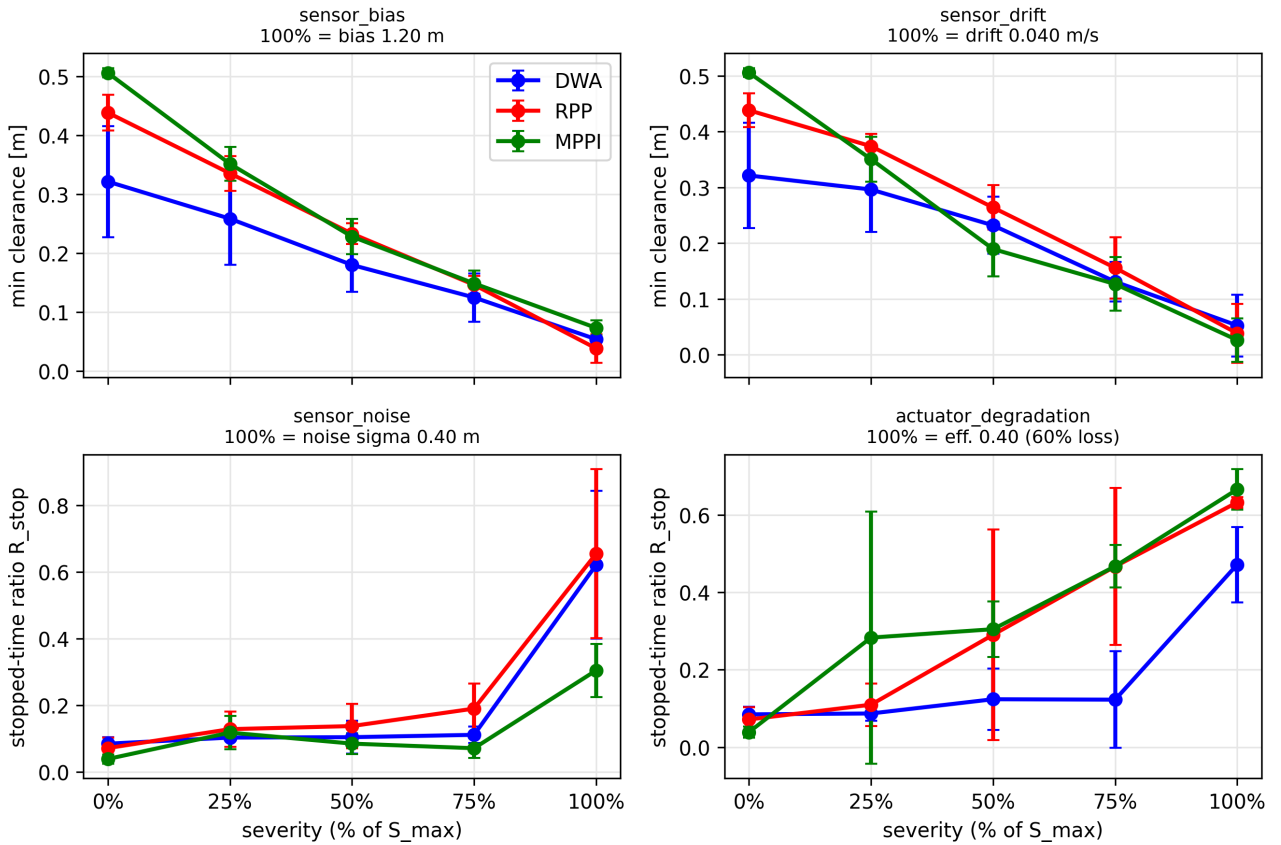
H1 - primary-metric dose-response (mean $\pm$ std)

Figure 1: Dose-response degradation on primary family metrics (mean $\pm$ std). Perception faults (top) erode clearance; liveness faults (bottom) increase the stopped-time ratio.

5.3 H2: Controller Robustness Differentiation

We next ask whether the three planners really differ in robustness, comparing their SPL for each fault. Fig. 2 pools all active severities into one box per planner, and the pattern is immediate: for bias and drift the three boxes sit at almost the same height, and under noise MPPI sits only somewhat lower while all three planners collapse together at 100%, so the planners are essentially interchangeable under perception faults. Statistically, ten of the twelve pairwise comparisons are significant (Wilcoxon signed-rank paired test, Holm-corrected), but the two that are not are exactly DWA versus RPP under bias and noise. One caution: even where a perception-fault comparison is significant, the SPL gap mostly reflects a different driving style (a longer path lowers SPL even when the robot still reaches the goal), not a real robustness gap.

The picture changes completely under actuator degradation. In Fig. 2 the MPPI box drops far below DWA and RPP, and Fig. 3 shows why, severity by severity: the three curves, which stay close together for every perception fault, here fan out in a clear order of collapse. MPPI falls first, losing SPL from the very first severity level and reaching zero by 75%; RPP degrades step by step from 50% and reaches zero at 100%; DWA degrades the most gracefully, still around 0.35 SPL at full severity. This order is the robustness ranking: DWA leads MPPI by a median $+0.24$ SPL (a large effect, Cliff's $\delta = 0.68$), RPP sits in between ($+0.037$ SPL over MPPI, $\delta = 0.37$), and DWA and RPP are indistinguishable at low severity and separate only at the top levels (overall median $+0.006$ SPL, $\delta = 0.17$). In short, a shared perception fault is an equalizer, whereas a fault in the actuator penalizes the model-based planner (MPPI) far more than the short-horizon sampler (DWA) or the model-free tracker (RPP).

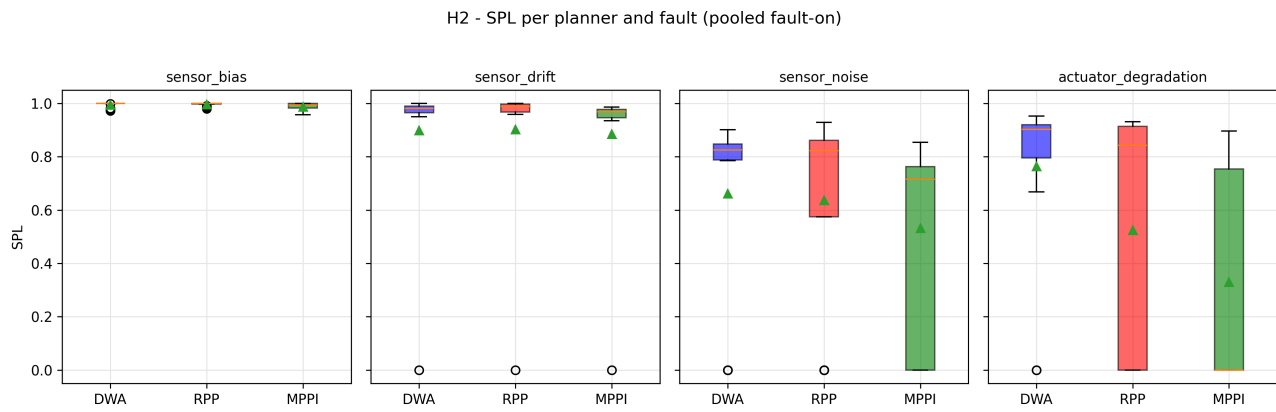


Figure 2: SPL distributions by planner across fault types (pooled over active severities). Overlapping boxes (bias, drift, and largely noise) indicate shared perception degradation; separated boxes (actuator) highlight controller-specific robustness.

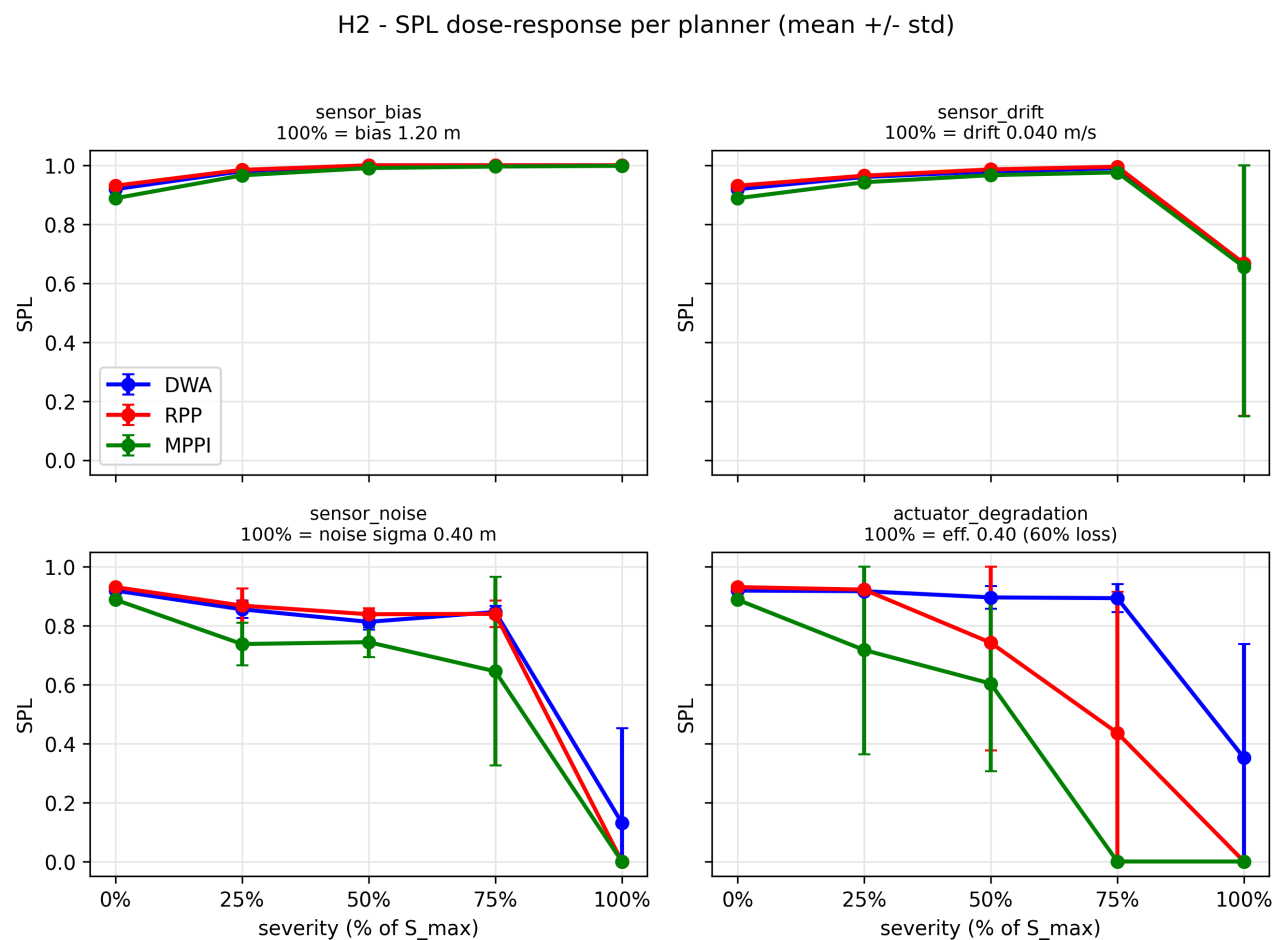


Figure 3: Dose-response SPL across fault severities. Under actuator degradation the planners separate progressively from 25% severity onward, with MPPI at zero from 75%.

5.4 Qualitative Failure Signatures

Beyond statistical metrics, spatial trajectory analysis (Fig. 4) exposes the four distinct qualitative failure signatures anticipated by H3:

- **Sensor Bias (Equalizer):** Yields a 100% SR with zero collisions. Planners maintain their trajectory styles but systematically erode safety margins.
- **Sensor Drift (Late Collision):** Empirically, all 6 recorded collisions across the 306 trials occurred exclusively at 100% drift severity. Notably, all of these collisions struck the internal obstacle at coordinate $(-3, 3)$ (the final

tight passage before the goal) while maintaining a safe ~ 1.5 m gap from the perimeter walls. The growing LiDAR ranges make the robot “cut the corner” into the obstacle.

- **Sensor Noise (Freeze / Liveness Collapse):** Triggers zero collisions but a sharp rise in R_{stop} . At maximum severity the three planners collapse (success rate 0–0.17): rather than cleanly recovering, the robot fails to complete the task. It either wanders with inflated path lengths (up to ~ 15 m against a ~ 10 m optimum), freezes in place, or stalls just short of the goal tolerance, with hyper-conservative, oscillatory behavior.
- **Actuator Degradation (Outcome Differentiator):** Causes entrapment with zero collisions. While DWA and RPP attempt to persist near the goal, MPPI surrenders entirely in open space, often halting ~ 8 m away from the target (Fig. 4(b)).

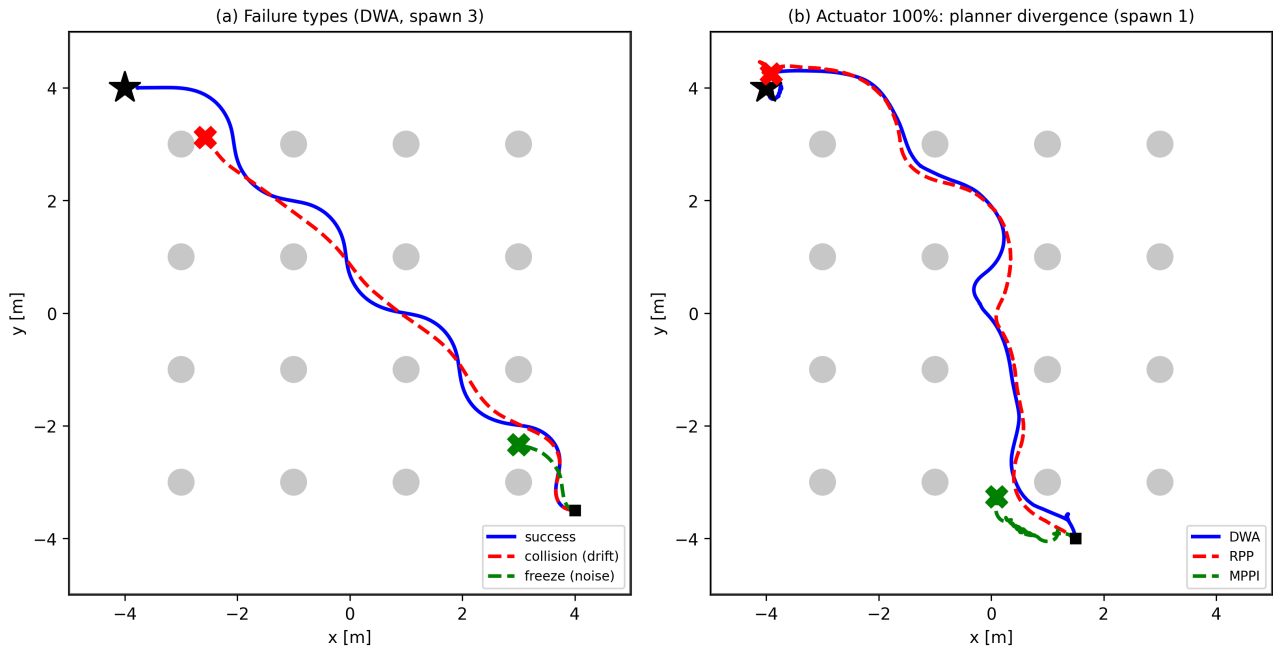


Figure 4: Qualitative trajectory analysis. (a) Failure types under a fixed planner (DWA, spawn 3): a successful nominal run, a late collision under drift (X near the internal obstacle at $(-3, 3)$), and freezing under noise. (b) Planner divergence under actuator degradation at 100% (spawn 1): DWA reaches the goal and RPP is entrapped on it, while MPPI surrenders in open space ~ 8 m short of the target.

6 Discussion and Limitations

6.1 Mechanistic Interpretation of Planner Robustness

The main finding of this study is simple: how much the three planners differ in robustness depends on how deep the fault reaches into the navigation stack. Under the perception faults (bias, drift, noise) the planners are almost indistinguishable in outcome, because they share the same perception-to-costmap pipeline. That pipeline acts as a bottleneck: it hands every controller the same corrupted view of the world, so their different control strategies never get the chance to matter.

Actuator degradation is different: it bypasses perception and corrupts the motion commands directly, so the control strategies are finally exposed. This confirms our hypothesis that the planners become more distinguishable the closer a fault acts to the controller. They separate as $DWA \geq RPP \gg MPPI$, and the ranking is best explained by how much each planner relies on an internal model of the robot. MPPI is the most sophisticated of the three: a model-predictive controller that samples many control sequences, rolls them out through a kinematic model of the vehicle, and combines them by cost-weighted averaging [4]. Precisely this dependence on a model makes it the most fragile here: a per-wheel loss of effectiveness is a large model-mismatch, so MPPI’s roll-outs no longer match reality and it abandons the task in open space.

DWA and RPP, in contrast, carry little or no model that the fault can invalidate. RPP is a purely geometric path tracker for which “vehicle dynamics are not modeled” [5], and DWA reacts to the world through short-horizon velocity sampling against the current costmap [4]; neither predicts far ahead with a dynamics model, so both absorb the actuator mismatch far more gracefully. Taken together, these results show that planner robustness is strongly stressor-specific, extending earlier Nav2 benchmarks that stressed mainly perception and the environment [2] to the actuator side.

6.2 Methodological and Statistical Limitations

Our design has clear limitations. Perfect localization isolates reactive avoidance cleanly, but it is unrealistic: in a real robot a LiDAR fault would also corrupt the AMCL pose estimate, which would likely make failures happen sooner. All the experiments also ran in a single simulated Gazebo world, with the actuator degradation fixed to the right wheel.

The statistics carry caveats too. With only $n = 6$ trials per cell, the Wilcoxon signed-rank test runs at its practical floor, so it has limited power and a non-significant result should not be read as no effect. Pooling the four active severities to $n = 24$ for the H2 comparisons buys power but adds some pseudo-replication, since severities that share a spawn are not fully independent; the large effect sizes (Cliff's δ) make us confident this does not drive the conclusions. Because a failed episode gives an SPL of exactly zero, the many resulting ties also pushed the test from its exact p -value to a normal approximation.

A larger campaign would help along two separate axes. More repetitions per cell would raise the power where the effects are small, which is exactly where our two non-significant comparisons sit (DWA versus RPP under bias and noise), and a finer severity grid would locate the failure threshold more precisely, since under drift and noise almost all of the collapse happens between the 75% and 100% levels. We kept the grid at four severities by six spawns because running the full campaign is already time-consuming.

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